ADITYA UNIVERSITY

Mern Stack Project

Final Review

Student Management System

Presented by

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Presentation Outline

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ABSTRACT

- This project focuses on developing a **Student Management System** to efficiently manage and organize student information using web technologies. The system helps in maintaining accurate records of student details, improving data accessibility, and reducing manual workload.

- The system includes attributes such as **student name, roll number, date of joining, year of study, branch, personal details, parent details, attendance records, fee details, and hostel status (hosteller/non-hosteller)**, along with online fee payment tracking.

- Data handling techniques such as **form validation, structured data storage, and error handling** are applied to ensure accuracy and consistency. The system allows easy entry, updating, and retrieval of student information.

- The application is developed using **HTML, CSS, JavaScript for frontend**, and **PHP/Node.js with MySQL/MongoDB for backend and database management**. It supports functionalities like **student registration, attendance management, fee tracking, and report generation**.

- Overall, the system provides an efficient and user-friendly solution for managing student data digitally and can be further enhanced with advanced features like authentication, analytics, and mobile support.

System Description

The system developed in this project manages and organizes student-related information, including academic, personal, attendance, and financial details in a structured and efficient way.

System Details

- System Type: Web-based application
- Platform: Browser-based (accessible through Chrome, Edge, etc.)
- Architecture: Client–Server Model
- Database Type: Structured database (MySQL / MongoDB)

Modules Included

- Student Registration Module
- Student Details Management
- Parent Details Module
- Attendance Management
- Fee Management & Online Payment
- Hostel / Non-Hosteller Management

System Description

Attributes Used

- student_id – Unique identifier for each student
- name – Student full name
- roll_number – Unique roll number
- date_of_joining – Admission date
- year_of_study – Current academic year
- branch – Department (e.g., CSE, ECE)
- student_details – Address, contact information
- parent_details – Parent name, contact details
- attendance – Attendance percentage/records
- fee_details – Total fees, paid amount, due amount
- payment_mode – Online / Offline payment
- hostel_status – Hosteller / Non-Hosteller

Target Functionality

- Add new student records
- Update and delete existing records
- Track attendance details
- Manage fee payment and online transactions
- View and search student information easily

System Output

The system provides organized student data such as: Complete student profile Attendance reports ,Fee payment status, Hostel allocation details .

Problem Statement

In many educational institutions, managing student information manually has become a complex and time-consuming task. Maintaining records such as student details, attendance, and fee payments using traditional methods leads to errors, data loss, and difficulty in accessing information.

Main Problem:
To develop a **centralized and efficient Student Management System** that can accurately store, manage, and retrieve student data, including academic, personal, attendance, and financial details using a digital platform.

Sub-Problems:
1. Handling large volumes of student data
2. Managing incomplete or incorrect data entries
3. Maintaining accurate attendance records
4. Tracking fee payments and online transactions
5. Ensuring quick and easy access to student information

Data Preprocessing

Data Cleaning

Data cleaning is performed to handle **missing, inconsistent, or incorrect student data**. Missing values in fields such as attendance, fee details, or contact information are corrected or filled using appropriate methods to ensure data accuracy and reliability.

Data Integration

Data integration involves combining data from different modules such as **student details, parent details, attendance records, and fee information** into a single structured system. This ensures that all student-related data is available in one centralized database for easy access and management.

Data Reduction

Data reduction is used to remove **redundant or unnecessary information** from the system. Only important attributes like **student ID, name, roll number, attendance, fee details, and branch** are retained to improve system efficiency and performance.

Data Transformation

Data transformation converts data into a suitable format for processing.
For example, attendance can be converted into **percentage format**, and fee details can be categorized as **Paid, Pending, or Overdue** to make the system more understandable and efficient.

Methodologies

The project follows a **structured web application development approach** to manage student data efficiently using frontend, backend, and database technologies.

Frontend Development

- Developed using **HTML, CSS, and JavaScript**
- Provides user interface for entering and viewing student details
- Ensures responsive and user-friendly design

Backend Development

- Implemented using **PHP / Node.js**
- Handles server-side logic and request processing
- Manages data communication between frontend and database

Database Management

- Uses **MySQL / MongoDB** for storing student records
- Stores data such as student details, attendance, fees, and parent information
- Ensures data consistency and easy retrieval

System Working Flow

- User enters data through frontend interface
- Request is sent to the server
- Backend processes the request
- Data is stored/retrieved from the database
- Output is displayed to the user

➡ **Flow:** User → Server → Database → Output

Modules

MODULES OVERVIEW:

•The system starts with the **Logo Page**, which represents the identity of the application.

•The **Login Page** allows existing users to securely access the system using their credentials.

•The **Register Page** is used by new users to create an account.

•After login, users are directed to the **Student Dashboard**, which provides an overview and navigation to a
modules.

•The **Student Details Module** stores and manages student personal and academic information.

•The **Attendance Module** is used to record and monitor student attendance.

•The **Academics Module** maintains subject details, marks, and performance records.

•The **Fee Details Module** manages fee information, payment status, and online payments.

•The **Logout Module** ensures users can safely exit the system.

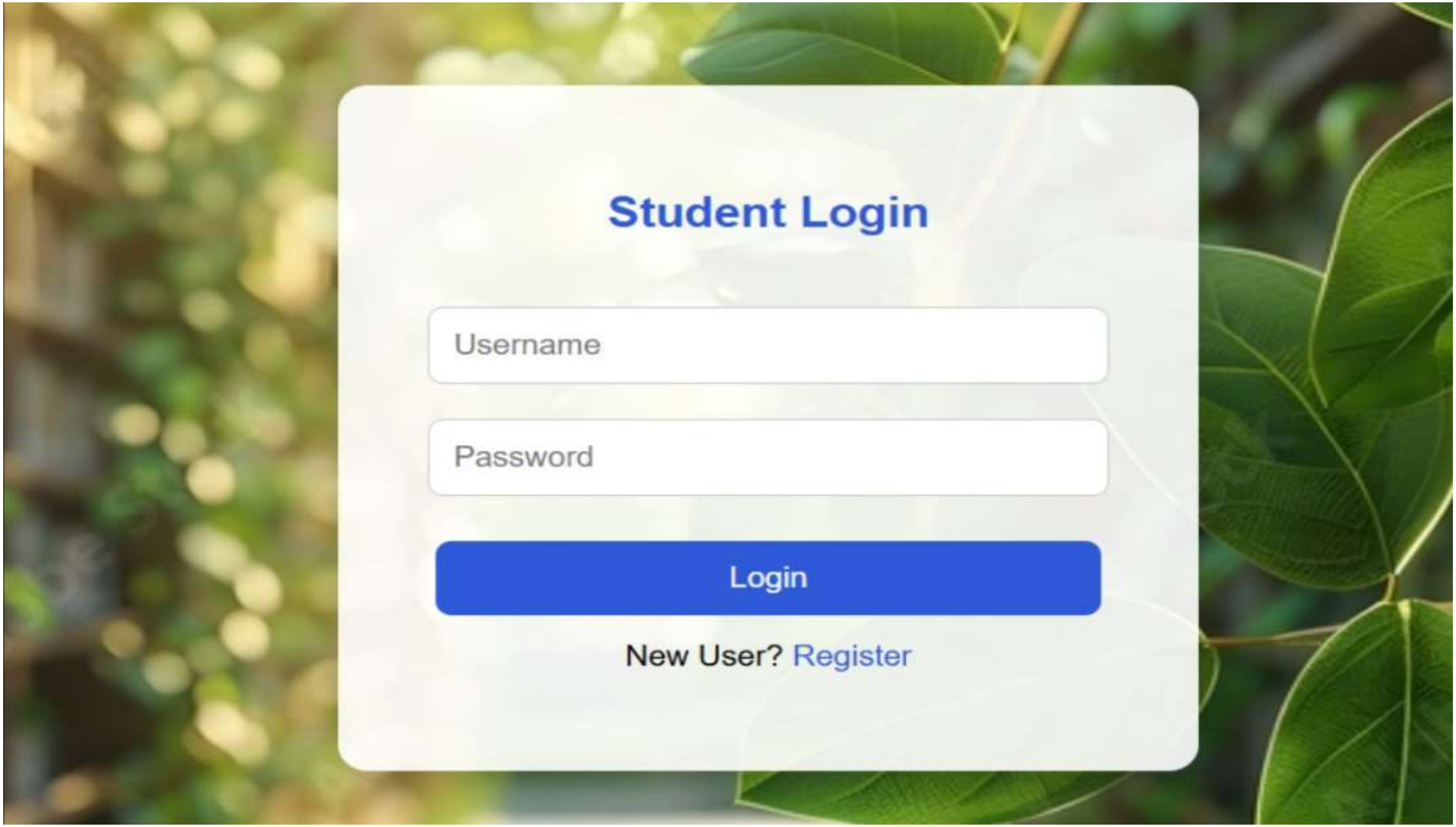
Modules

Logo Page:
Displays the system logo and provides an entry point to the application.

Modules

Login Page

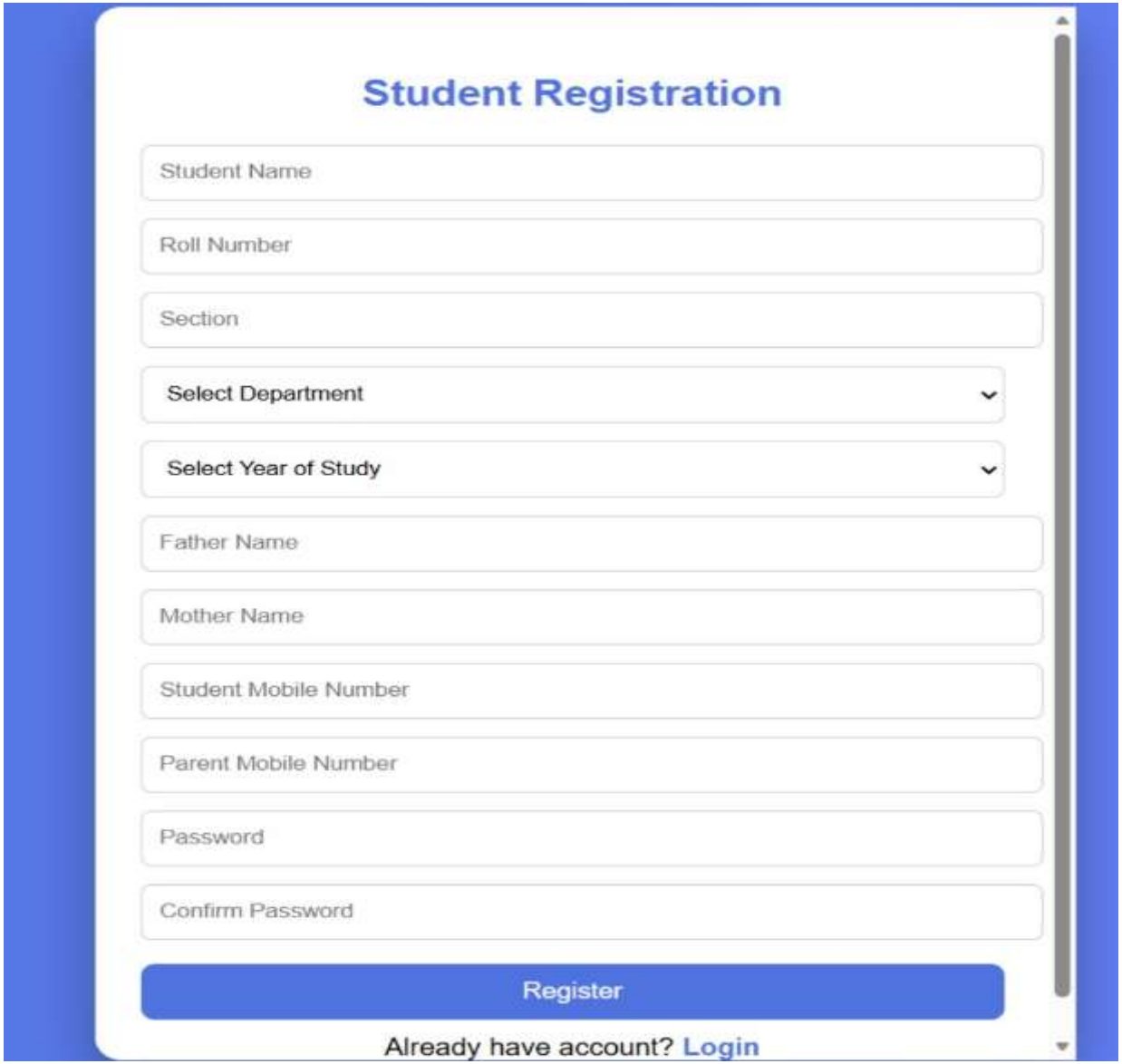
Allows users to securely log in using their credentials.



Modules

Register Page

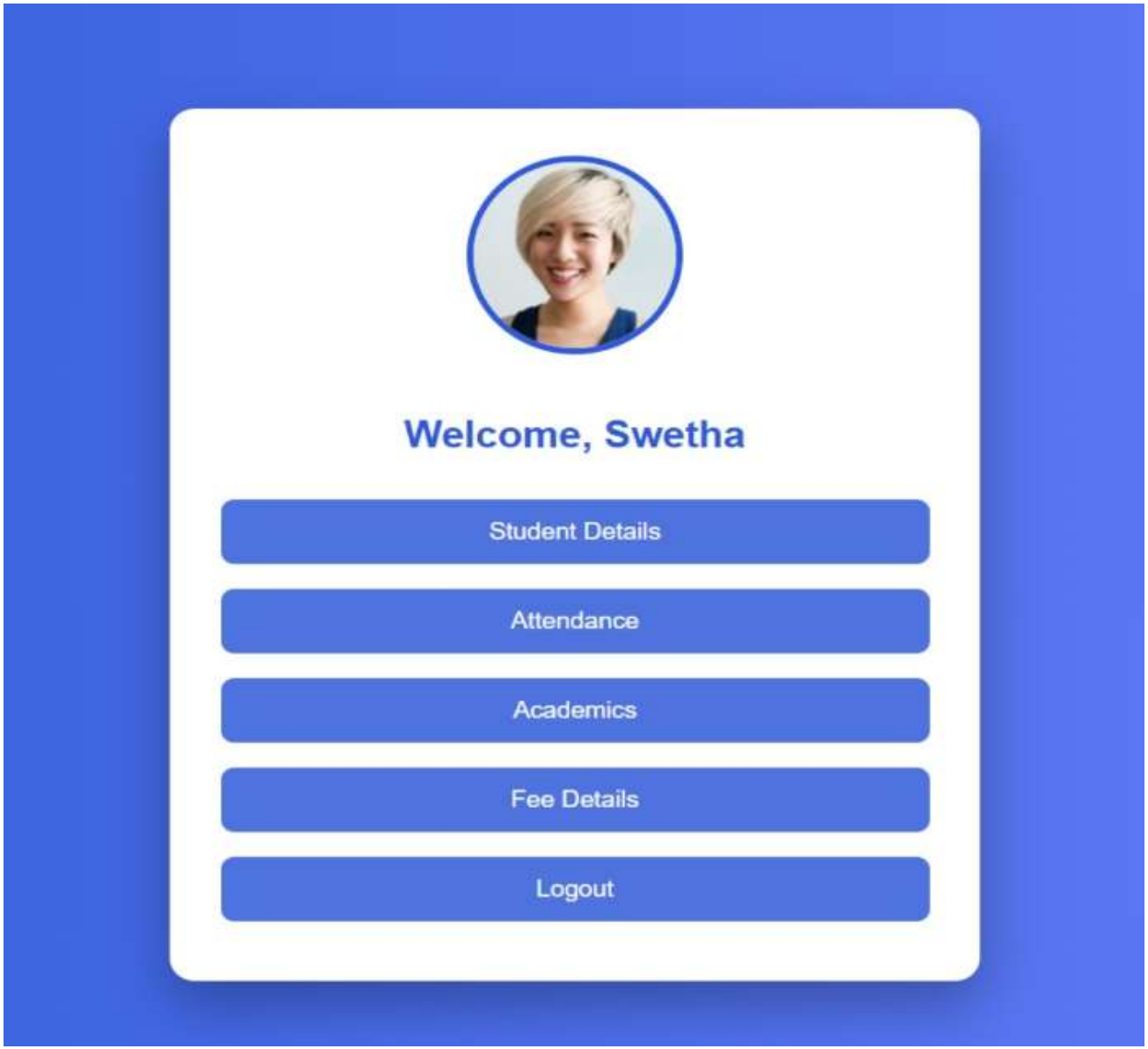
Enables new users to create an account in the system.



Modules

Student Dashboard

Provides an overview of student information and quick access to all modules.



Modules

Student Details

Stores and displays personal and academic information of students.

Student Details

Name: Swetha

Father Name: Santhosh

Mother Name: Padma

Roll No: 24B11AI087

Year of Joining: 2024

Currently Pursuing: B.Tech 2nd Year

Branch: AIML (AI)

Section: A

Student Mobile: 9876543210

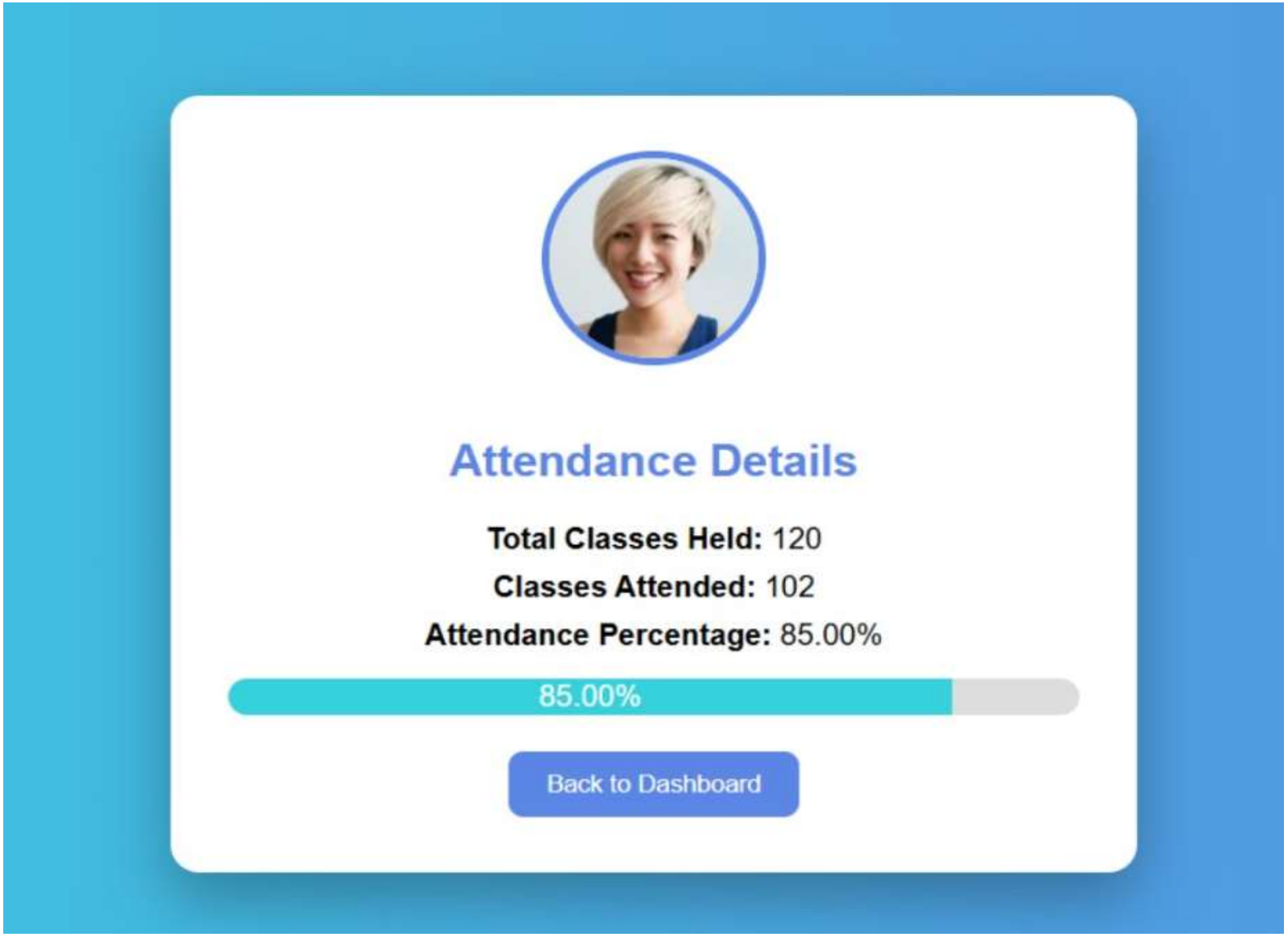
Parent Mobile: 9123456780

Back to Dashboard

Modules

Attendance Details

Tracks and manages student attendance records.



Modules

Academics Details

Maintains academic performance, subjects, and results of students.

Academic Details

Subject	Marks	Status
Mathematics	85	Pass
Data Structures	72	Pass
Operating Systems	65	Pass
DBMS	40	Fail
AI Fundamentals	90	Pass

Total Passed: 4

Total Failed: 1

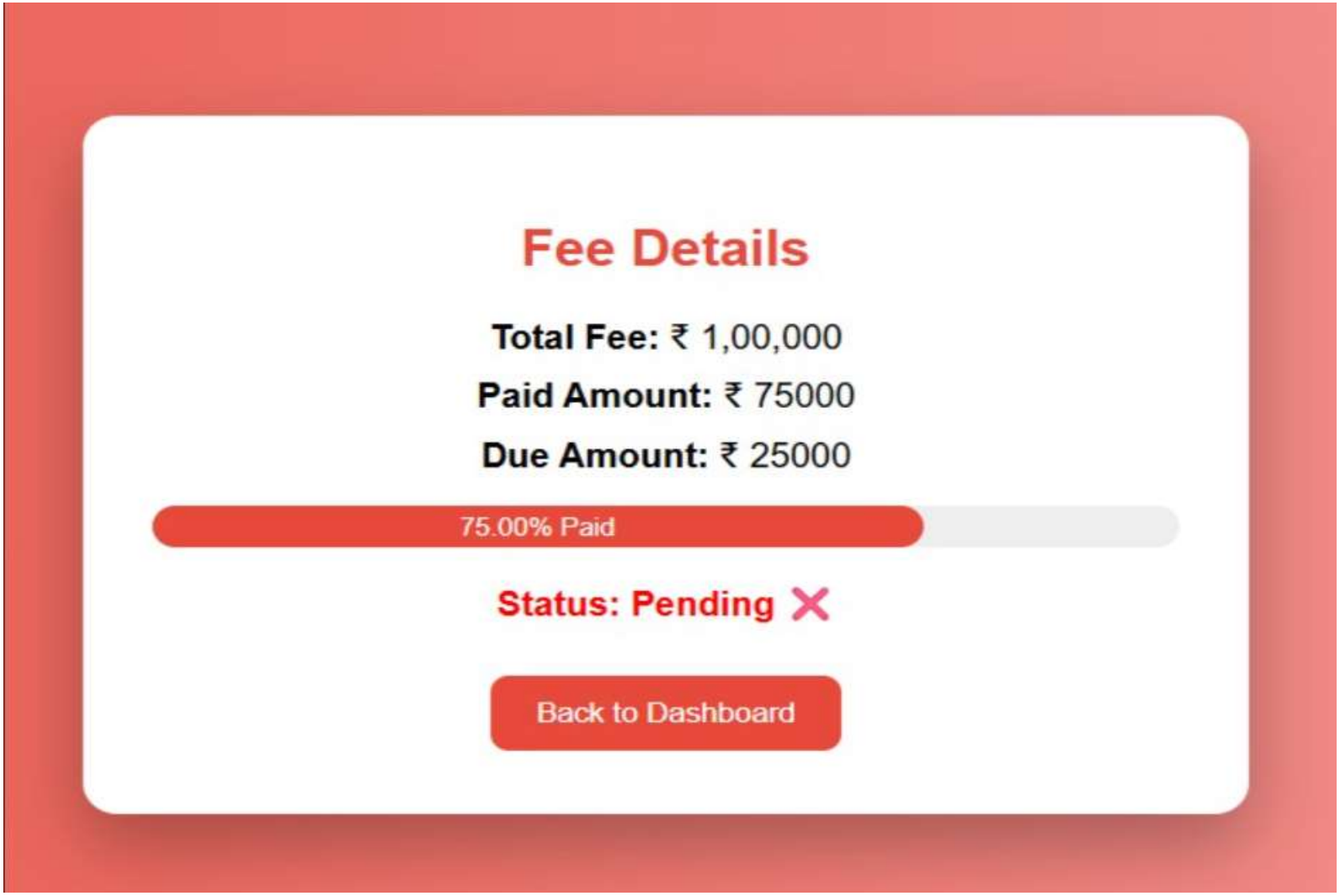
CGPA: 7.04

Back to Dashboard

Modules

Fee Details

Manages fee structure, payment status, and online transactions.



System Features / Implementation

- **Student Record Management**
  The system allows users to add new student details, update existing information, and delete records when required, ensuring accurate and up-to-date data.

- **Attendance Tracking**
  It enables efficient recording and monitoring of student attendance, helping maintain proper attendance history and reports.
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- **Fee Management System**
  The system manages fee details, including total fees, paid amount, and pending dues, along with tracking online payment status.
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- **Search & Filter Functionality**
  Users can quickly search and filter student data based on roll number, name, or branch for easy access and management.

- **User-Friendly Interface**
  The application provides a simple, intuitive, and easy-to-use interface, allowing users to interact with the system without difficulty.

Conclusion

- **Project Summary**
  The Student Management System is a web-based application developed to manage student information in a structured and efficient manner. It integrates various modules such as student details, attendance tracking, academic records, and fee management into a single centralized platform, making data handling simple and organized.

- **Advantages of the System**
  The system significantly reduces manual paperwork and human errors, improves data accuracy, and saves time for administrators and faculty. It provides quick and easy access to student records, enhances data organization, and offers a user-friendly interface for smooth operation.

- **Future Enhancements**
  In the future, the system can be enhanced by developing a mobile application for better accessibility, implementing advanced security features to protect data, automating processes like attendance and fee notifications, and integrating analytics for better decision-making.

THANK YOU